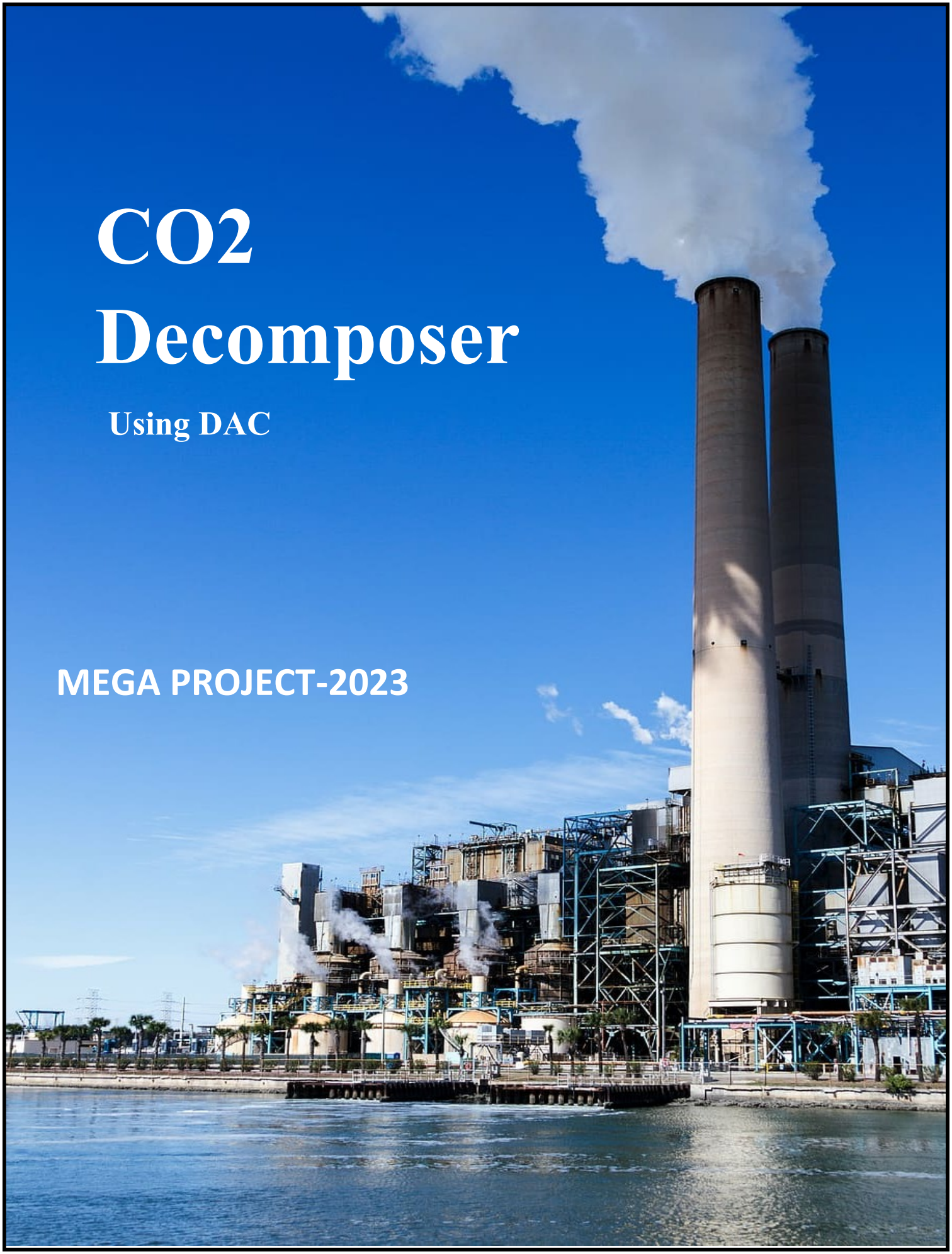
CO2 Decomposer

Using DAC

MEGA PROJECT-2023



Direct Air Capture and Storage Technology

A Synopsis on

Submitted by

ARCHIT RAGHAVENDRA KULKARNI: 2122010119

VAIBHAV RAMESH KASHID: 2021000347

BHOOMY DURGESH LINGRAS: 2021000722

SHREYAS AMRUT DESAI: 2021000732

In partial fulfilment for the award of the degree of

Electronics and Telecommunication

Under the guidance of

Prof. Atul Nigvekar

THE FACULTY OF MEGA PROJECT

Department of Electronics and Telecommunication Engineering

KIT's College of engineering, Kolhapur

2023-24

INDEX

CHAPTER NO.	TITLE	Page No.
1	PROBLEM STATEMENT	4
2	RELEVANCE	5
3	LITERATURE SURVEY	7
4	PRESENT THEORIES AND PRACTICES	10
5	OBJECTIVES AND SCOPE OF PROJECT	12
6	RESEARCH METHODOLOGY	15
7	PROJECT REQUIREMENTS	17
8	REFERENCES	19

Problem Statement

The rising levels of atmospheric carbon dioxide (CO_2) pose a critical threat to global climate stability. Current efforts to reduce CO_2 emissions from industrial processes and transportation are insufficient to meet climate targets. Currently, the Direct Air Capture (DAC) technology offers a promising solution but faces significant challenges in terms of cost-effectiveness, energy efficiency and scalability. The challenge is to optimize room-scale DAC technology via an open-source approach, iteratively improving DAC systems with crowd-sourced ideas and an open data set for experimentation. We all are familiar with the potential consequences of CO_2 therefore our goal is to enhance carbon dioxide capture efficiency, promoting sustainability and addressing climate change.

Relevance

What is DAC?

Direct Air Capture (DAC) is an advanced technology designed to remove CO_2 directly from the Earth's atmosphere. It plays a crucial role in addressing climate change by actively reducing the concentration of CO_2 (a major greenhouse gas responsible for global warming). DAC systems employ specialized chemical processes or absorbent materials to capture CO_2 from the air. Once captured, the CO_2 can be stored underground or utilized in various applications, such as in carbon-neutral fuels or carbon removal products. DAC offers a potential solution to combat climate change by helping to balance CO_2 emissions and achieve net-zero or even negative emissions, thus contributing to a more sustainable and climate-resilient future.

The need of DAC.

The relevance of a project on DAC lies in its potential to address the urgent and escalating global climate crisis. With atmospheric CO2 levels reaching alarming concentrations, DAC offers a vital strategy to actively remove excess CO2 from the air, helping to mitigate climate change's adverse effects. DAC technology has the potential to complement traditional emissions reduction efforts and create a pathway towards achieving carbon neutrality. By developing cost-effective, energy-efficient, and scalable DAC solution, this project holds the promise of contributing significantly to global efforts to combat climate change and safeguard the future of our planet for generations to come.

Literature Survey

The review of literature provides a comprehensive overview of relevant studies and research on the present theory and practices followed in DAC-CCS technology. This literature review highlights the key findings and insights from various sources, demonstrating the significance and potential impact of above mentioned technology in addressing the challenges faced by present practitioners.

Source 1: ” CARBON DIOXIDE CAPTURE AND STORAGE (2005)” * Intergovernmental Panel on Climate Change (IPCC) of WMO (World Meteorological Organization)*

Findings: Special Report which describes sources, capture, transport, and storage of CO2. It also discusses the costs, economic potential, and societal issues of the technology, including public perception and regulatory aspects.

Source 2: “A Study on the Evolution of Carbon Capture and Storage Technology Based on Knowledge Mapping” * Hong-Hua Qiu & Lu-Ge Liu writers, Article Published by MDPI (May2018)*

Findings: Detailed Information about the topics includes carbon emissions and environmental protection, research and development activities and social practical issues, meanwhile, the main emerging trends include emerging techniques and processes, emerging materials, evaluation of technological performance and socioeconomic analysis in the research area of CCS technology.

Source 3: “Carbon Capture and Storage in Developing Countries: A Perspective on Barriers to Deployment”* by Natalia Kulichenko & Eleanor Ereira A WORLD BANK STUDY*.

Findings: Project Finance for Power Plants with Carbon Capture and Storage in Developing Countries (Ch-6, pg 80-106)

Source 4: "Direct Capture of CO2 from Ambient Air" * By- Eloy S. Sanz-Perez, Christopher R. Murdock, Stephanie A. Didas and Christopher W. Jones* of School of Chemical & Biomolecular Engineering, Georgia Institute of Technology, 311 Ferst Drive, Atlanta, Georgia 30332-0100, United States and Department of Chemical and Environmental Technology, ESCET, Rey Juan Carlos University, C/Tulipan s/n, 28933 Mo´ stoles,´ Madrid, Spain*

Findings: A historical overview of the field of DAC and exhaustive description of the use of chemical sorbents targeted at this application, includes solvents and solid sorbents that interact strongly with CO2 are described, including basic solvents, supported amine and ammonium materials, and metal–organic frameworks (MOFs), as the primary classes of chemical sorbents.

Conclusion: The review of literature underscores the importance and relevant studies and research on the present theory and practices followed in DAC-CCS technology. It showcases how DAC-CCS technology can make a significant positive impact on the environment which contribute to sustainability goals. These findings serve as a strong foundation for the development and implementation of DAC-CCS technology at room scale

Present Theories and Practices

Carbon dioxide (CO2) emissions are a major driver of global climate change and represent a pressing environmental challenge. This overview explores current theories and practices up to September 2021 concerning the prevention and reduction of CO2 emissions, encompassing various sectors, strategies, and policy frameworks.

1. Carbon Capture and Storage (CCS):

Theory: Involves capturing CO2 emissions at the source and storing them underground.
Practice: Implementation of CCS technologies in power plants and industrial facilities to prevent CO2 from entering the atmosphere.

2. Processes for Reducing CO2 Emissions:

Despite all the efforts aimed at minimizing CO2 emissions, tens of billions of tons are still released every year. Thus, CO2 capture was proposed by the IPCC as a technology necessary to restrain the growth of the atmospheric CO2 concentration. Conventional CO2 capture has commonly addressed CO2 emissions from large stationary sources, such as fossil-fuelbased power stations, cement plants, oil refineries, and iron or steel industry installations, as examples. According to the different processes where carbon capture can be implemented, three main technologies are described.

2.1. **Precombustion**: These technologies are typically used when starting from natural gas or syngas (CO and H2). Several processes can be used to create suitable feeds for energy generation as well as CO2 separation, including steam reforming (reaction 1), as well as gasification of pulverized coal (IGCC), coke, or oil residues (reaction 2). The reaction between CO and water, the water gas shift reaction, yields a gas stream composed of primarily CO2 and hydrogen (reaction 3). The removal of CO2 yields a high-purity H2 stream available for electricity production. CO2 capture is relatively easy in these cases because its concentration ranges between 15 and 60%. On the other hand, the first steps of fuel preparation for coal gasification and steam reforming are challenging and expensive.

$$\mathrm{CxHy} + \mathrm{xH2O} \rightarrow \mathrm{xCO} + \left(\mathrm{x} + \frac{\mathrm{y}}{2}\right)\mathrm{H2} \quad (1)$$

$$\mathrm{CxHy} + \frac{\mathrm{x}}{2}\mathrm{O2} \rightarrow \mathrm{x\,CO} + \frac{\mathrm{y}}{2}\mathrm{H2} \quad (2)$$

$$\mathrm{CO} + \mathrm{H2O} \rightarrow \mathrm{CO2} + \mathrm{H2} \quad (3)$$

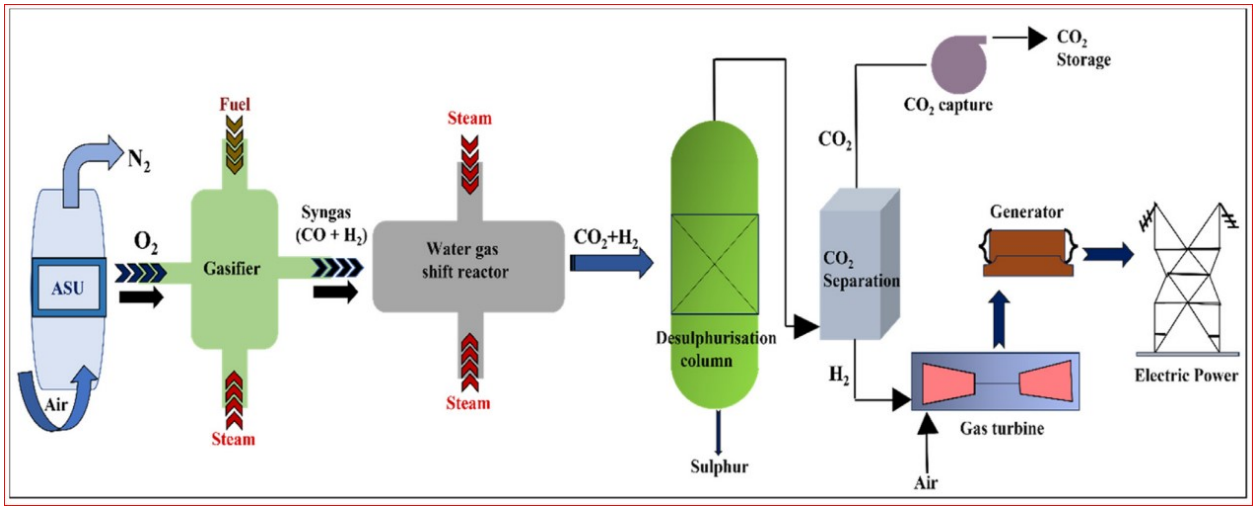


Fig. Depiction of pre-combustion CO2 capturing technique

2.2. **Capture During Combustion:** Two main technologies are used for capture during combustion, oxy-fuel combustion and carbonation−calcination. Oxy-fuel processes carry out the combustion with pure oxygen instead of air. A part of the combustion gases is typically recirculated in such processes. The flue gas volume is considerably reduced by the use of oxygen instead of air, and the CO2 concentration is concomitantly increased (up to 90%). However, these technologies are not yet highly favored because of the high cost of the air separation needed to obtain pure oxygen.

2.3. Postcombustion: CO2 can also be captured from flue gas released after combustion. A number of technologies can be used for this approach because existing combustion facilities do not need to be modified. Typically, the flue gas needs to be previously desulfurized to a great extent which is compulsory by law in most countries. The main disadvantage of this approach is the reduced CO2 concentration, which ranges from ∼4 vol % for a natural gas combined cycle processes to 13−15 vol % for a coal-fired combustion plant. Global average surface temperature change from 2006 to 2100 determined by multimodel simulations. All changes are relative to 1986−2005. Time series of projections and a measure of uncertainty (shading) are shown for scenarios RCP2.6 (blue) and RCP8.5 (red). The mean and associated uncertainties averaged over 2081−2100 are given for all RCP scenarios as colored vertical bars at the right-hand side of each panel. The number of Coupled Model Intercomparison Project Phase 5 (CMiP5) models used to calculate the multimodel mean is indicated. Reprinted with permission from ref 5. Copyright 2014 IPCC.

Chemical Reviews Review DOI: 10.1021/acs.chemrev.6b00173 Chem. Rev. 2016, 116, 11840−11876 11841 such as SO2 and NOx are also a problem for many separation processes; for example, because of their reactivity toward amines, which are the compounds commonly used for postcombustion CO2 capture, amine-based separation processes require flue gas pretreatment to significantly reduce the concentrations of these strongly acidic gases. To date, large-scale postcombustion CO2 separation processes have been mainly based on liquid amine absorption techniques. Aqueous solutions of monoethanolamine (MEA), diethanolamine (DEA), and methyldiethanolamine (MDEA) have been successfully used to capture carbon dioxide on a large scale. Despite their proven efficacy in capturing CO2, absorption processes still entail major drawbacks.

The most important problems are related to the corrosive properties of liquid amines, the amine losses incurred during the operation, and, significantly, the energy consumption in the regeneration process. Therefore, in recent years many efforts and significant funding have been focused on the development of alternative processes for CO2 capture, such as cryogenic, adsorption, or membrane-based technologies. Carbonation−calcination uses the carbonation reaction between quicklime (calcium oxide, CaO) and CO2 to produce limestone (calcium carbonate, CaCO3).Calcium carbonate is then introduced in a calcination reactor, where the opposite process is performed. Calcination yields regenerated quicklime and a gas stream enriched in CO2. The main advantages of this technology are the low cost of the initial limestone used and its reactivity toward both CO2 and SO2. Thus, no desulfurization step is needed prior to carbonation−calcination. However, the high temperature used during the carbonation (650 °C) and calcination reactions (900 °C) as well as material attrition are a significant economic drawback. The carbonation step is commonly performed with flue gas, in a postcombustion CO2 capture setup. Nevertheless, it can also be applied to precombustion, for example, during coal or biomass gasification (Table 1).

	syngas (precombustion)	**oxy-fuel combustion**	**flue gas (postcombustion)**
pressure (atm)	>5	>50	1
temperature (°C)	>100	<50	<100
CO2 concentration (vol %)	~35	>90	4-14

Table 1: Typical Pressures, Temperatures, and CO2 Concentrations of the Different Gas Streams Used in CCS Technologies

1.3. Reduction of CO2 Emissions vs Negative Carbon Technologies Given the slow realization that the last century of carbon dioxide emissions has significantly impacted the climate, society has continued to use fossil fuels at an ever-expanding rate over the past few decades. Although the IPCC recognized the need for rapid and immediate reductions in fossil fuel use in 1990, the rate of anthropogenic CO2 emissions has continued to expand rapidly, with little slowdown in sight. While the use of noncarbon energy technologies is quickly growing, the growth does not match the pace of a growing global population's energy demands. As such, there appears to be little probability of a slowing of CO2 emissions, globally, in the near future. In fact, the latest IPCC report from Working Group III suggests that society has reached a point where it is very likely that stabilizing the atmosphere at twice the preindustrial level will require negative CO2 emissions in the later part of this century, as can be seen in Figure 2.

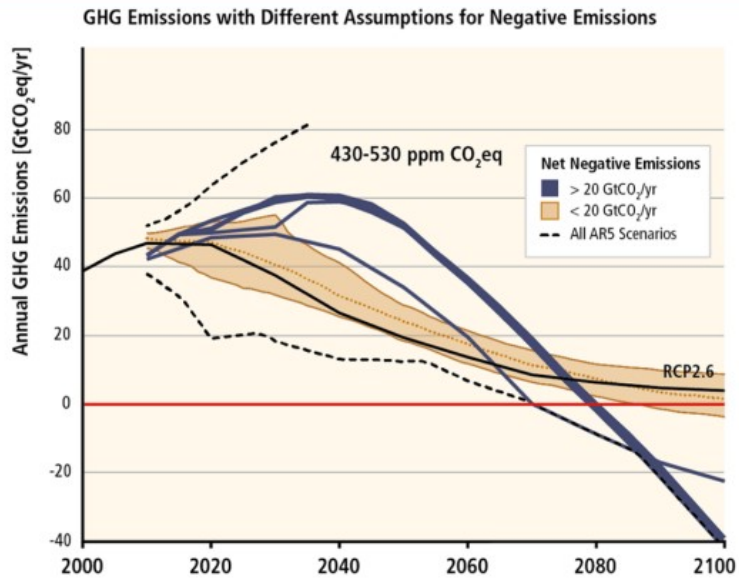


Figure 2. Development of total GHG emissions for scenarios reaching ~450 to ~500 (430–530) ppm of CO_2eq in 2100 with and without net negative CO_2 emissions larger than 20 Gt of CO_2/year. Ranges are given for the 10th–90th percentile of scenarios. All approaches that extend below the red line require substantial negative emissions. Reprinted with permission from ref 21. Copyright 2014 Cambridge University Press.

To this end, the recent climate talks in Paris have identified the need for rapid deployment of "negative carbon technologies," technologies that can remove CO2 from the atmosphere. Additionally, the report from the National Research Council (NRC) on Climate Intervention also concluded that research and development of negative carbon technology is warranted. This is in contrast to the conventional CO2 capture technologies described above, which, even if widely deployed, would only slow the rate of increase of the global atmospheric CO2 concentration.

This Review describes the advances made in the direct extraction of CO2 from ambient air using sorbents, or direct air capture (DAC), as one of the few potential “negative carbon” technologies. The history of DAC from a climate perspective is first presented, followed by the identification of an array of sorbents and processes that have been examined for their utility in extracting CO2 from the atmosphere. A brief discussion of nonsorbent-based approaches to DAC is also included. Finally, an overview of the feasibility and cost of DAC processes is presented.

Conclusion: These theories and practices are often interconnected and implemented simultaneously to create a comprehensive approach to CO2 emissions reduction. It's important to note that developments in this field continue, and new strategies and policies are emerging frequently.

Basic block diagram:

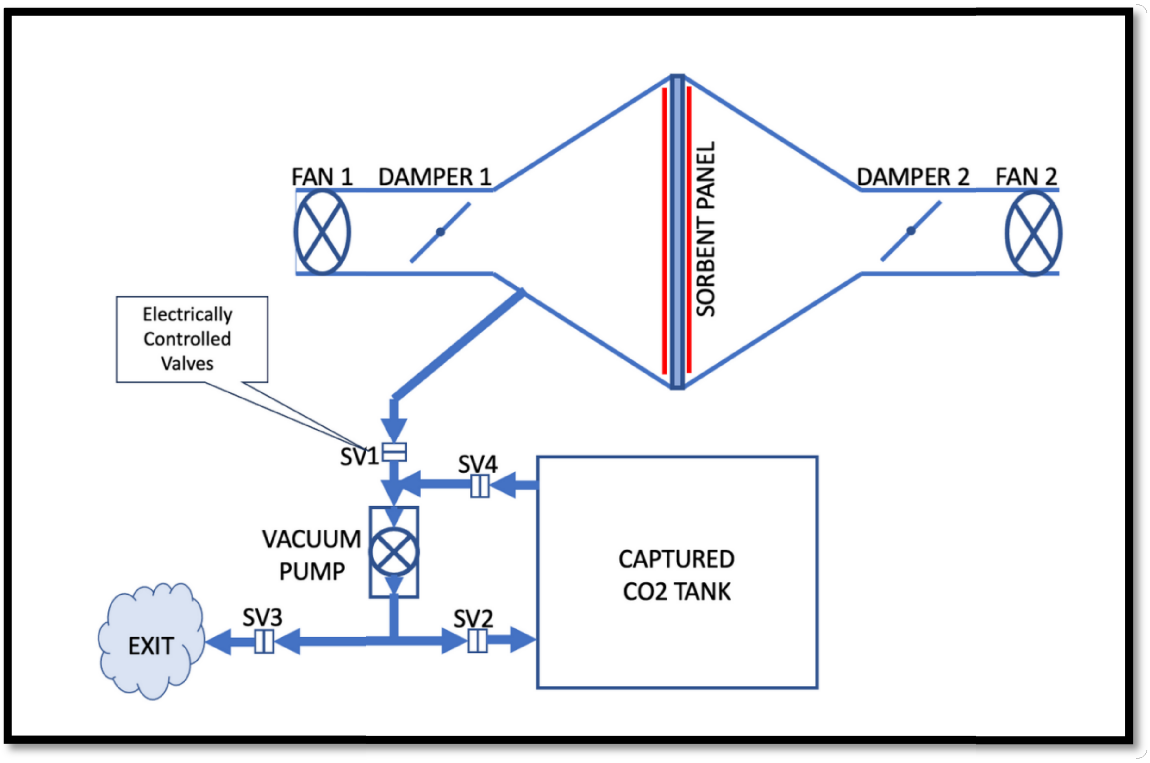


Figure: Schematic representation of direct air capture of CO2 with utilization and storage.
Source: Redrawn from David Sandalow et al. (2018).

Objectives

The objective of this project is to design, develop, and implement an innovative Direct Air Capture (DAC) system that addresses the urgent need for scalable and sustainable carbon removal solutions. This project aims to contribute to the mitigation of climate change by achieving the following key objectives:

1. Development of an Efficient DAC Technology:

- Design and construct a state-of-the-art DAC system that maximizes CO2 capture efficiency while minimizing energy consumption.
- Investigate and integrate advanced capture materials and methods, such as chemical sorbents, adsorption technologies, and electrochemical approaches, to enhance DAC system performance.

2. Scalability and Cost Reduction:

- Engineer the DAC system with scalability in mind, ensuring it can be deployed at a sizeable capacity to have a meaningful impact on CO2 removal.
- Explore innovative engineering and manufacturing processes to reduce the overall cost of DAC technology, making it economically viable for widespread deployment.

3. Carbon Utilization and Storage:

- Investigate opportunities for the utilization of captured CO2, such as in the production of carbon-neutral fuels, construction materials, or other value-added products.
- Implement effective and environmentally responsible strategies for the secure and long-term storage of captured CO2, including geologic storage and utilization in carbon-negative applications.

4. Environmental Sustainability:

- Conduct a comprehensive life cycle assessment (LCA) to evaluate the environmental footprint of the DAC system, taking into account emissions associated with construction, operation, and decommissioning.
- Develop strategies to minimize water usage, land footprint, and other potential environmental impacts associated with DAC deployment.

Scope of the project

1. Direct Air Capture (DAC):

- **Scope:**
  - DAC technology involves capturing CO2 directly from the air using various chemical processes. It is seen as a potential solution for reducing atmospheric carbon dioxide levels.
  - The scope of DAC extends to applications where direct removal of CO2 from the atmosphere is necessary to achieve climate targets.
- **Challenges**:
  - High costs have been a significant challenge for DAC technologies. Implementing large-scale DAC facilities requires substantial investment.
  - Energy requirements for DAC processes can be considerable, and finding sustainable and low-cost energy sources is crucial.
- **Recent Developments:**
  - Ongoing research and development aim to improve the efficiency and reduce the costs associated with DAC technologies.
  - Pilot projects and demonstration plants have been initiated to test and scale up DAC capabilities.

2. Carbon Capture and Storage (CCS):

- **Scope**:
  - CCS involves capturing CO2 emissions produced from industrial processes or power generation and storing it underground to prevent it from entering the atmosphere.
  - The scope of CCS is broad and includes applications in power plants, industrial facilities, and other large-scale point sources of CO2 emissions.
- **Challenges**:
  - One of the main challenges for CCS is the cost of capturing and storing CO2. Implementation has been hindered by economic considerations and a lack of policy incentives.
  - Public acceptance and concerns about the long-term integrity of CO2 storage sites are challenges that need to be addressed.
- **Recent Developments:**
  - Some countries have started to invest in and implement CCS projects as part of their climate change mitigation strategies.
  - Advances in technology and increased awareness of the importance of carbon capture have led to the development of new and more efficient capture methods.

Integration and Future Outlook:

- **Combined Approaches:**
  - There is growing recognition that a combination of DAC and CCS technologies, along with other mitigation strategies, may be needed to achieve significant reductions in CO2 levels.

- **Policy and Funding:**
  - The future scope of DAC and CCS will likely be influenced by government policies, international agreements, and financial incentives that support the development and deployment of these technologies.

- **Technological Advances:**
  - Ongoing research and innovation are expected to drive improvements in both DAC and CCS technologies, making them more cost-effective and scalable.

Methodology

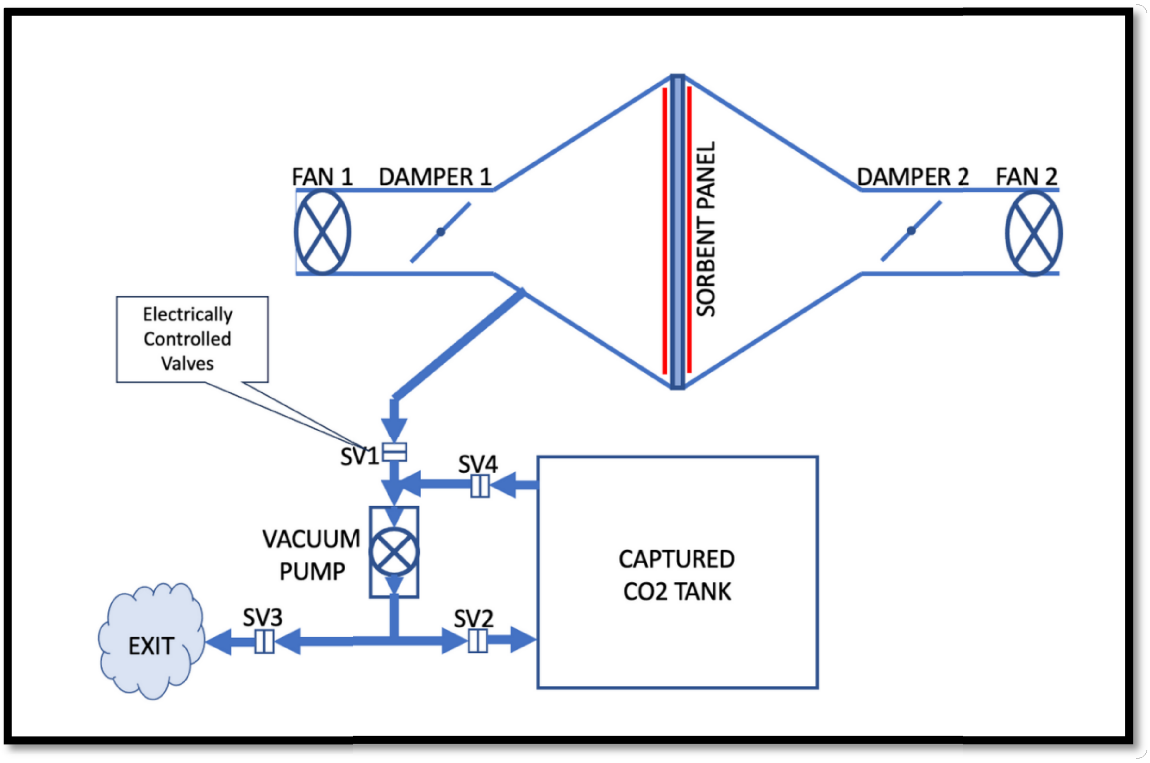


Figure: Schematic representation of direct air capture of CO2 with utilization and storage. Source: Redrawn from David Sandalow et al. (2018).

Working:-

1. Air Intake and Control:

- Fan 1 pulls in ambient air, and Dampers 1 regulate the airflow, directing it to the sorbent panel**.**

2. CO2 Absorption:

- The airflow, now containing CO2, passes through the sorbent panel where a specialized material selectively absorbs the CO2.

$$CxHy + xH2O \rightarrow xCO + \left(x + \frac{y}{2}\right) H2 \quad (1)$$

$$CxHy + \frac{x}{2} O2 \rightarrow x\,CO + \frac{y}{2} H2 \quad (2)$$

$$CO + H2O \rightarrow CO2 + H2 \quad (3)$$

3. CO2 Capture and Routing:

- Fan 2 is activated to guide the CO2-laden air toward the capture tank, controlled by Dampers 2 for optimized airflow.

4. CO2 Storage:

- Within the capture tank, CO2 is collected and stored for subsequent processing.

5. Vacuum-Assisted CO2 Extraction:

- The vacuum pump creates a low-pressure environment in the capture tank, facilitating the extraction of CO2 from the sorbent material.

6. Controlled Operation via Switches:

- The four switches play a vital role in managing the system, coordinating the fans, dampers, and vacuum pump at specific phases to ensure efficient CO2 capture.

In summary, the DAC system utilizes fans, dampers, a sorbent panel, a capture tank, a vacuum pump, and switches to draw in air, absorb CO2, store captured CO2, and facilitate its extraction. This systematic process showcases the potential of DAC in effectively reducing atmospheric CO2 levels, contributing to environmental sustainability.

Project Requirements

Damper:
An HVAC damper is a small valve or plate that regulates the airflow inside a building's heating, ventilation, and air conditioning (HVAC) system. The damper works by opening or closing to varying degrees in order to control the flow of air through different rooms or zones within the structure.

Sorbent Molecular sieves:
The most open molecular sieve sorbents, able to imbibe considerable amounts of larger molecules such as benzene and isopentane, were obtained using 1 N
and 2 N acid, but as acid strength increased further there was a decline in both crystallinity and sorption capacity until with 30 N acid an amorphous product resulted.

Sorbent Chamber - Inline fan:
Inline fans are mounted inline with the ductwork, bringing air from the inside and to the outside of the home. An inline fan may also be described as a remote-mounted fan. Inline fans are useful for applications that include: Ventilation of wet spaces such as bathrooms, kitchens and laundries.

Adafruit ESP32 Feather V2:
Adafruit Feather are a complete line of development boards from Adafruit that are both standalone and stackable. They're able to be powered by lipo batteries for on-the-go use or by their micro-USB plugs for stationary projects. Feathers are flexible, portable, and as light as their namesake.

Feather M0 Basic is a speedy chip mostly designed for Arduino programming. It can run CircuitPython but the board lacks the extra off-chip flash for extra storage space. Feather is the new development board from Adafruit, and like its namesake it is thin, light, and lets you fly!

While the RP2040 has lots of onboard RAM (264KB), it does not have built-in FLASH memory. Instead, that is provided by the external QSPI flash chip. On this board there is 8 MB, which is shared between the program it's running and any file storage used by MicroPython or CircuitPython.

Feather Display:

These displays are small, only about 1" diagonal, but very readable due to the high contrast of an OLED display. This screen is made of 128x32 individual white OLED pixels and because the display makes its own light, no backlight is required. This reduces the power required to run the OLED and is why the display has such high contrast; we really like this miniature display for its crispness! We also toss on a reset button and three mini tactile buttons called A B and C so you can add a mini user interface to your feather.

I2C multiplexer:

The TCA9548A I2C multiplexer allows you to communicate with up to 8 I2C devices with the same I2C bus. The multiplexer communicates with a microcontroller using the I2C communication protocol. Then, you can select which I2C bus on the multiplexer you want to address.

To address a specific port, you just need to send a single byte to the multiplexer with the desired output port number.

CO2 Sensor: (Sensirion) SCD-30 - NDIR CO2

The SCD30 from Sensirion is a high quality Nondispersive Infrared (NDIR) based CO_2 sensor capable of detecting 400 to 10000ppm with an accuracy of ±(30ppm+3%). In order to improve accuracy the SCD30 has temperature and humidity sensing built-in, as well as commands to set the current altitude. For additional accuracy the SCD30 also accepts ambient pressure readings!

Thermocouple:

thermocouple, also called thermal junction, thermoelectric thermometer, or thermal, a temperature-measuring device consisting of two wires of different metals joined at each end. One junction is placed where the temperature is to be measured, and the other is kept at a constant lower temperature.

References

sciencedirect.com

mdpi.com

pubs.acs.org

hackster.io

pubmed.ncbi.nlm.nih.gov/37269511/

pubmed.ncbi.nlm.nih.gov/28951132/

GROUP MEMBERS **SIGNATURES**

ARCHIT RAGHAVENDRA KULKARNI: 2122010119

VAIBHAV RAMESH KASHID: 2021000347

BHOOMY DURGESH LINGRAS: 2021000722

SHREYAS AMRUT DESAI: 2021000732

PROJECT GUIDE- Prof. ATUL NIGVEKAR

SIGNATURE: